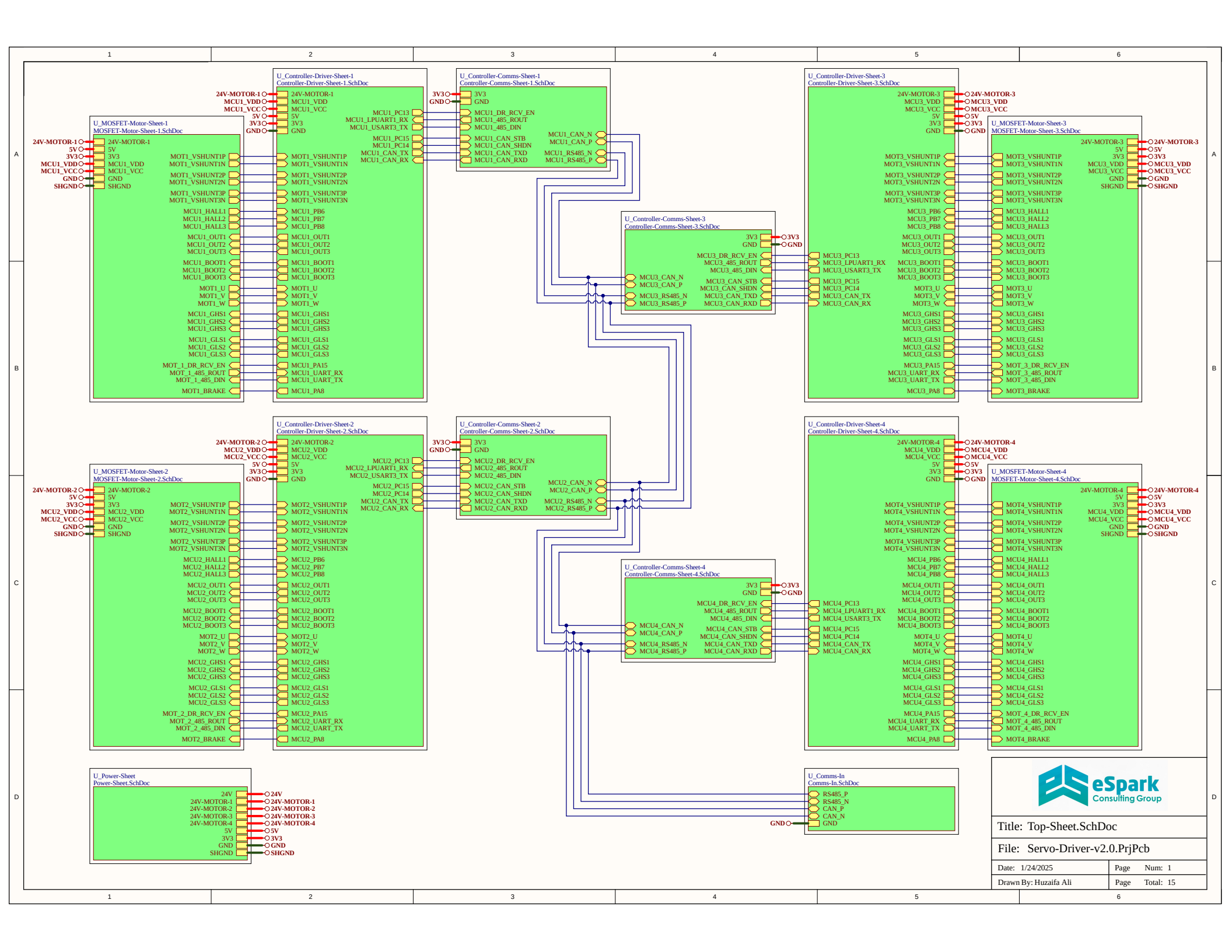
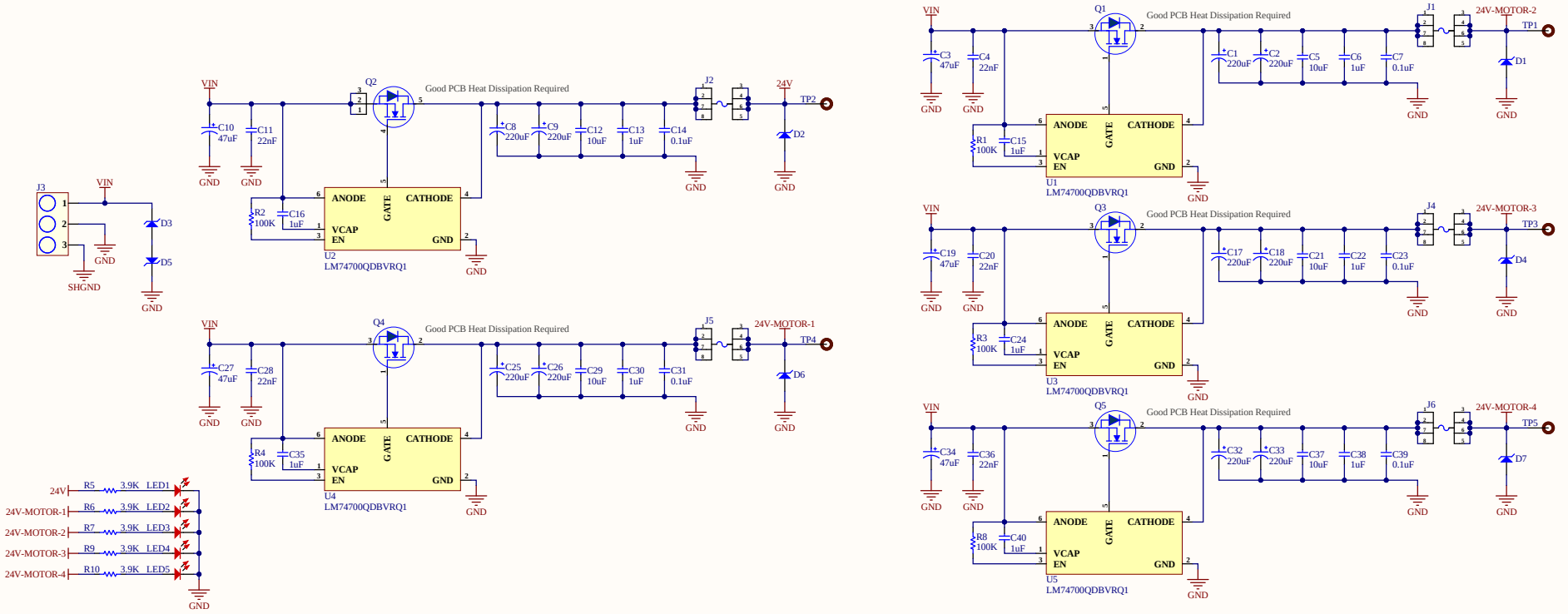
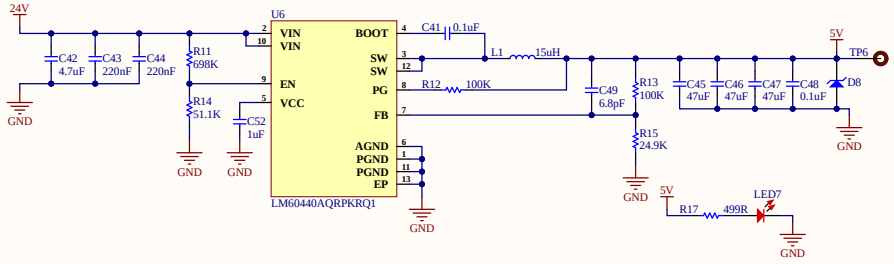

U_Controller-Driver-Sheet-1
Controller-Driver-Sheet-1.SchDoc
U_Controller-Comms-Sheet-1
Controller-Comms-Sheet-1.SchDoc
U_Controller-Driver-Sheet-3
Controller-Driver-Sheet-3.SchDoc
U_MOSFET-Motor-Sheet-1
MOSFET-Motor-Sheet-1.SchDoc
U_MOSFET-Motor-Sheet-3
MOSFET-Motor-Sheet-3.SchDoc
U_Controller-Comms-Sheet-3
Controller-Comms-Sheet-3.SchDoc
U_Controller-Driver-Sheet-2
Controller-Driver-Sheet-2.SchDoc
U_Controller-Comms-Sheet-2
Controller-Comms-Sheet-2.SchDoc
U_Controller-Driver-Sheet-4
Controller-Driver-Sheet-4.SchDoc
U_MOSFET-Motor-Sheet-2
MOSFET-Motor-Sheet-2.SchDoc
U_MOSFET-Motor-Sheet-4
MOSFET-Motor-Sheet-4.SchDoc
U_Controller-Comms-Sheet-4
Controller-Comms-Sheet-4.SchDoc
U_Power-Sheet
Power-Sheet.SchDoc
U_Comms-In
Comms-In.SchDoc
eSpark Consulting Group
Title: Top-Sheet.SchDoc
File: Servo-Driver-v2.0.PrjPcb
Date: 1/24/2025
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 15

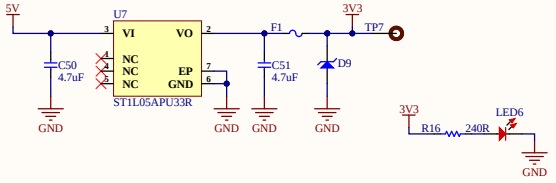
24V INPUT

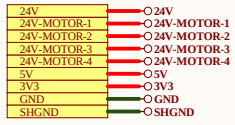


5V BUCK CONVERTER



3.3V LINEAR REGULATOR

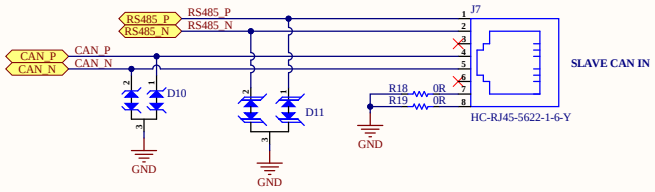




eSpark Consulting Group

Title: Power-Sheet.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 15

COMMUNICATION IN

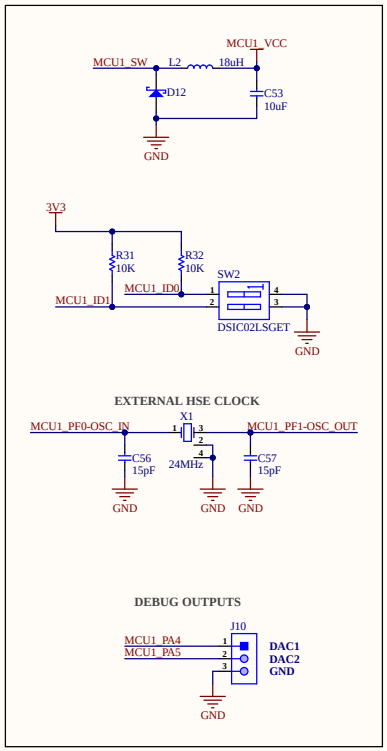


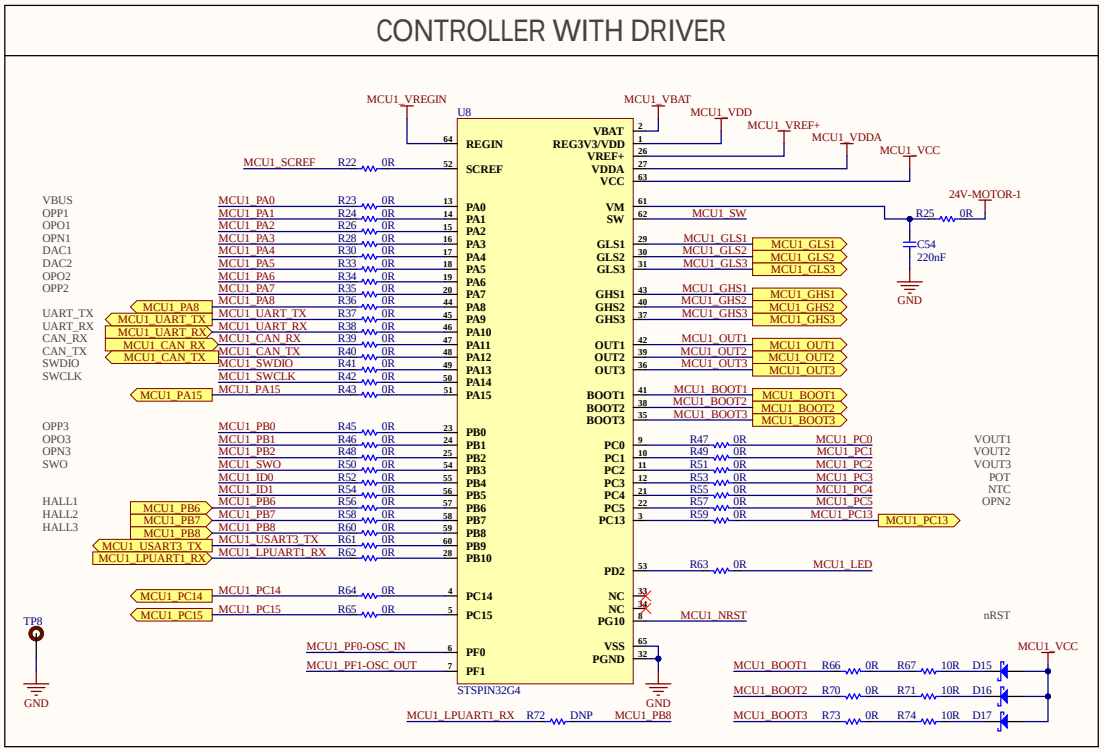
GND GND

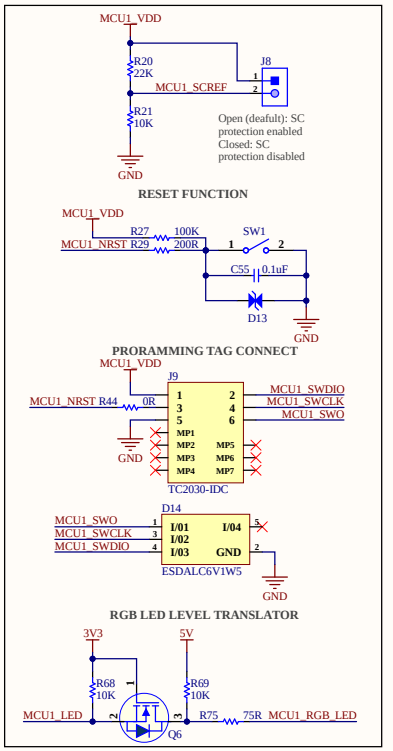
eSpark Consulting Group

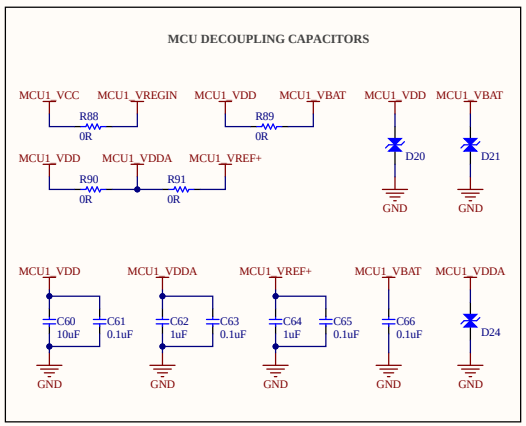
Title: Comms-In.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 15

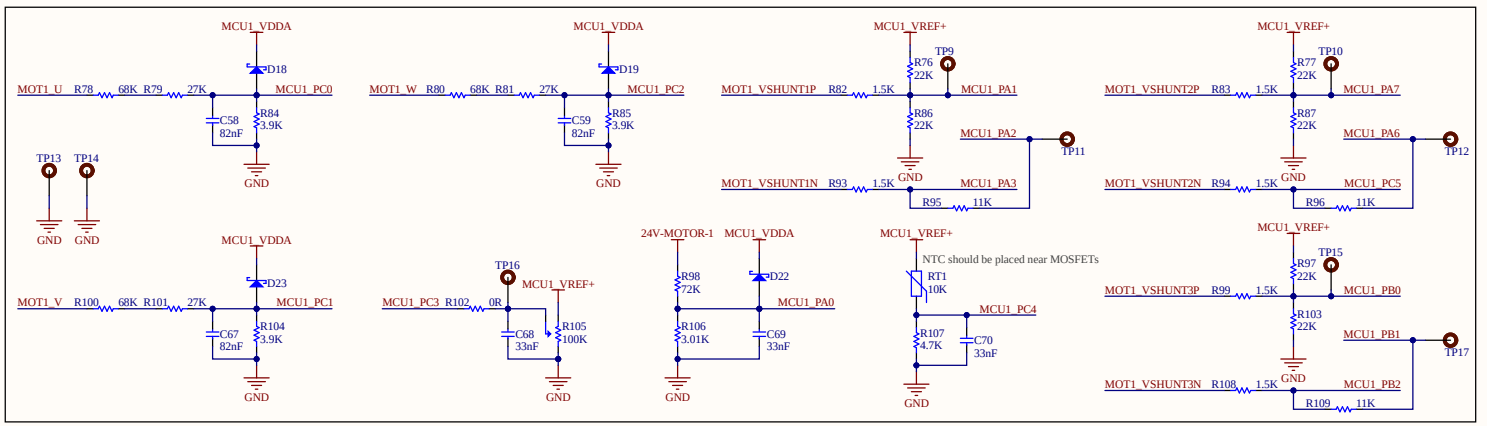
CONTROLLER WITH DRIVER

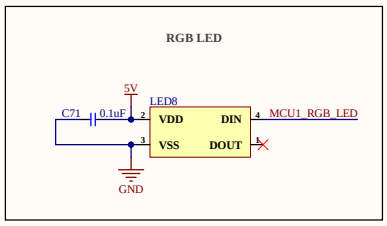












MOT1_VSHUNT1P — MOT1_VSHUNT1P
MOT1_VSHUNT1N — MOT1_VSHUNT1N
MOT1_VSHUNT2P — MOT1_VSHUNT2P
MOT1_VSHUNT2N — MOT1_VSHUNT2N
MOT1_VSHUNT3P — MOT1_VSHUNT3P
MOT1_VSHUNT3N — MOT1_VSHUNT3N

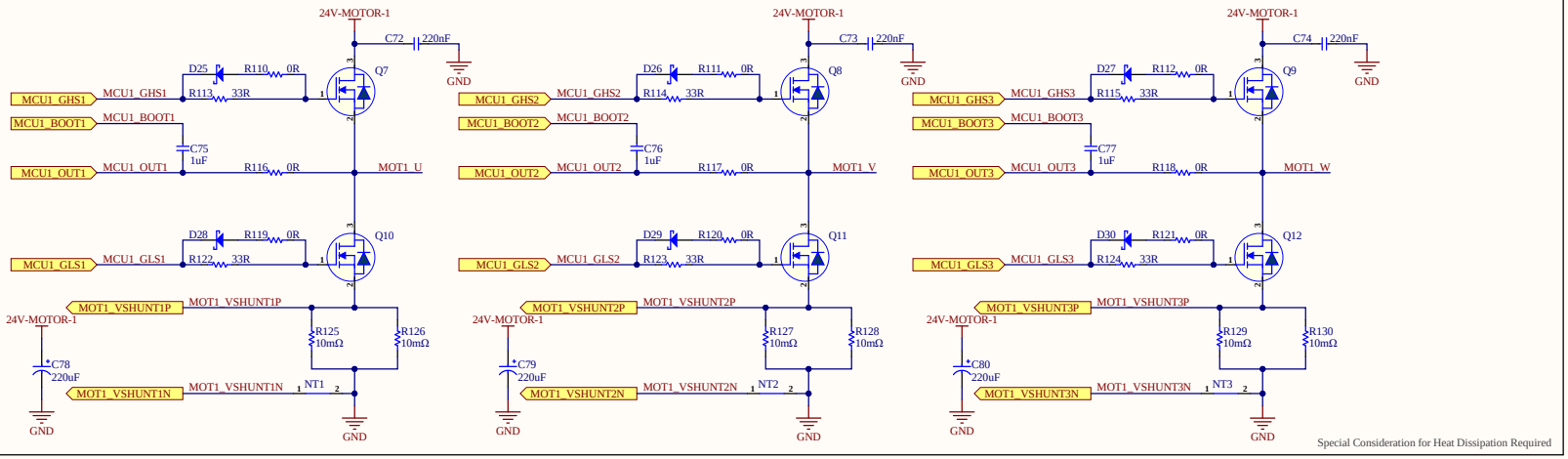
MOT1_U — MOT1_U
MOT1_V — MOT1_V
MOT1_W — MOT1_W

24V-MOTOR-1 — 24V-MOTOR-1
5V — 5V
3V3 — 3V3
MCU1_VDD — MCU1_VDD
MCU1_VCC — MCU1_VCC
GND — GND

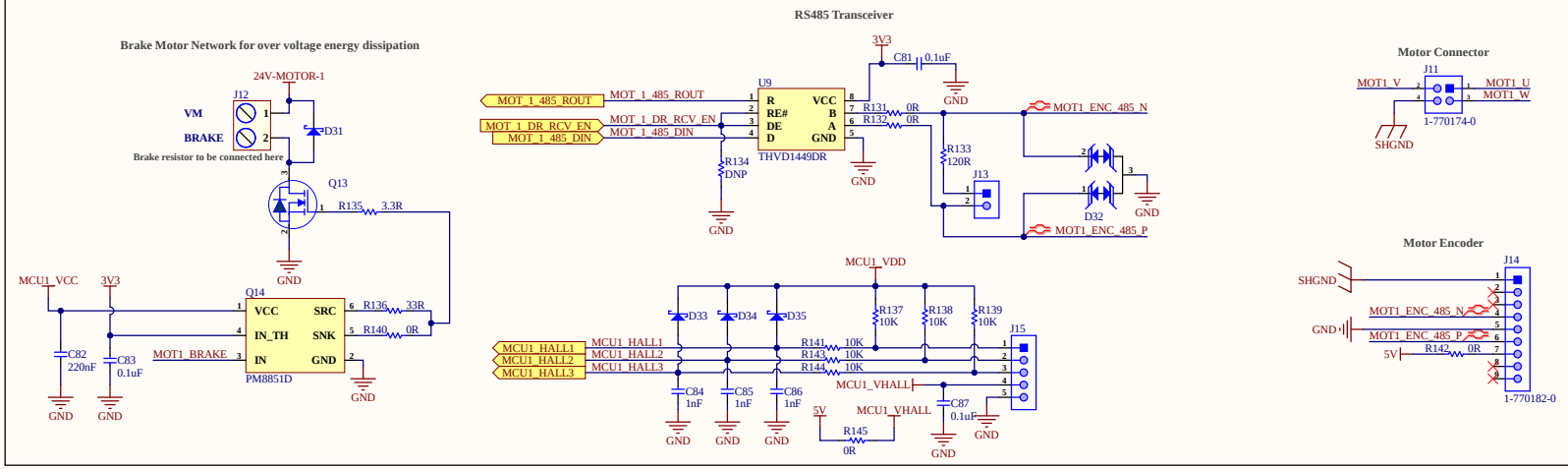
eSpark Consulting Group

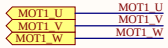
Title: Controller-Driver-Sheet-1.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS





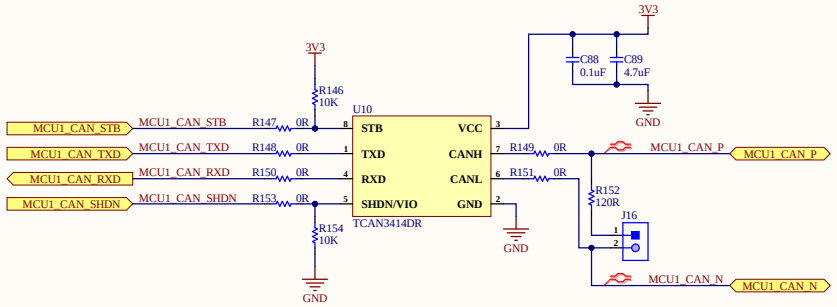




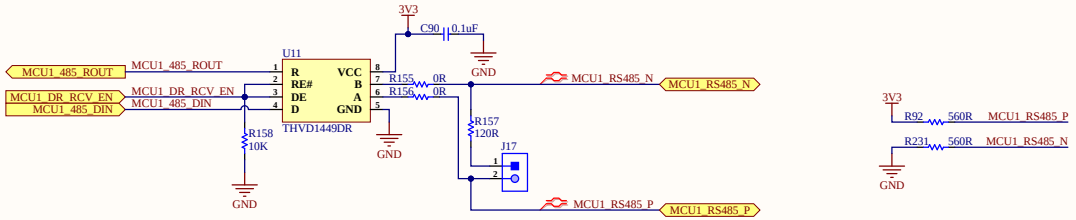
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-1.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485

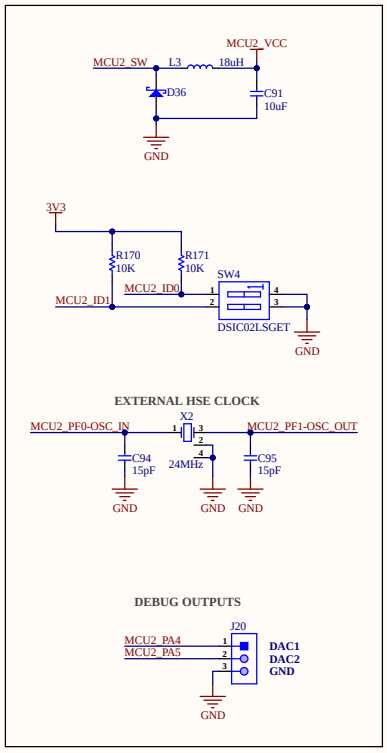


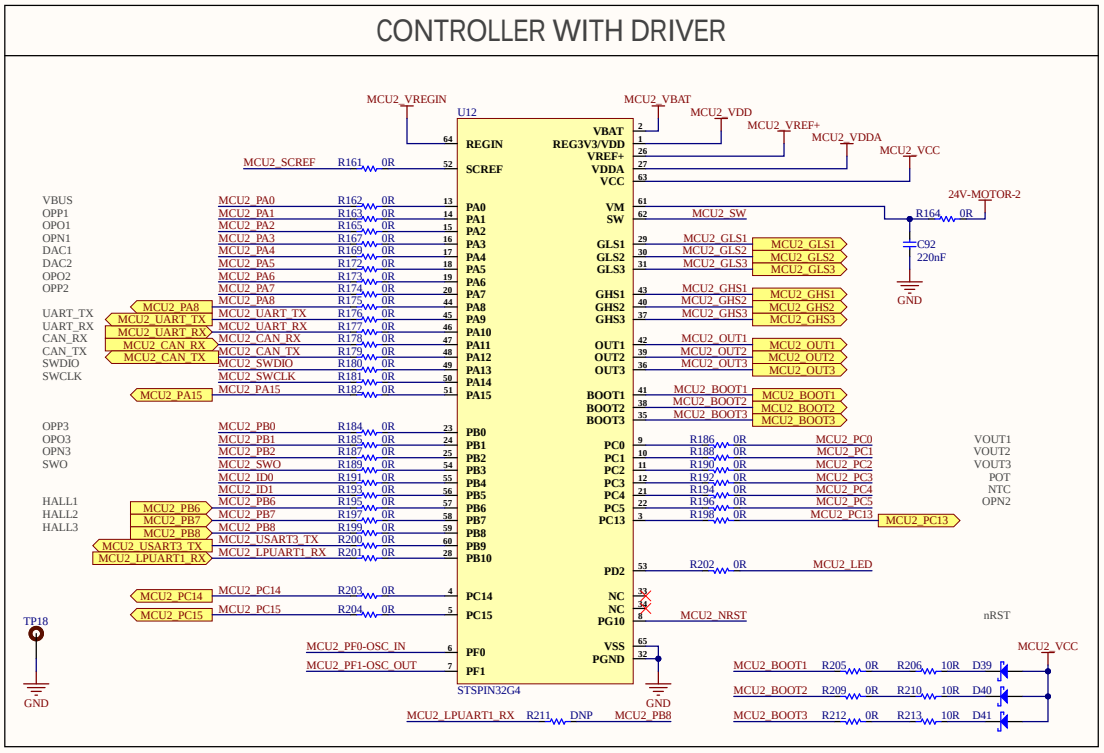


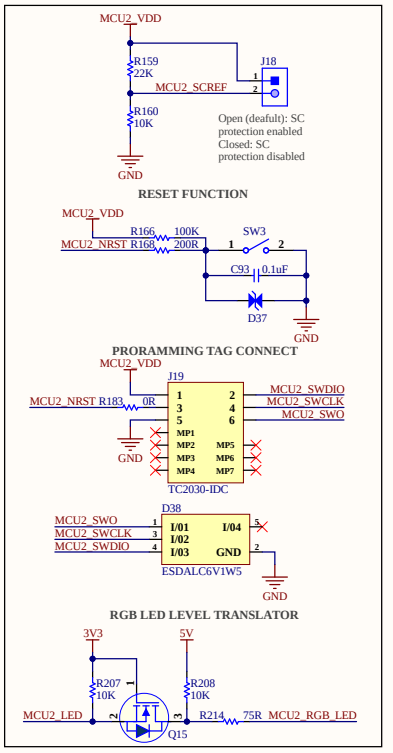
eSpark Consulting Group

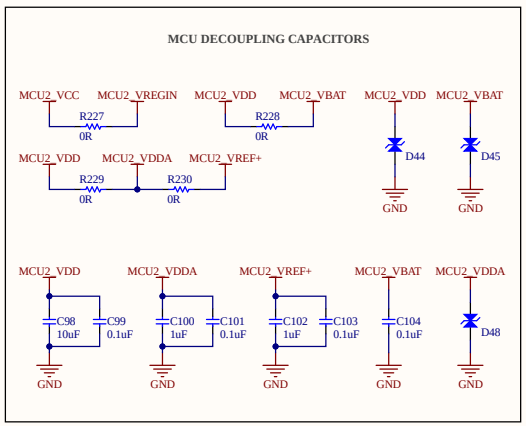
Title: Controller-Comms-Sheet-1.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/29/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 15

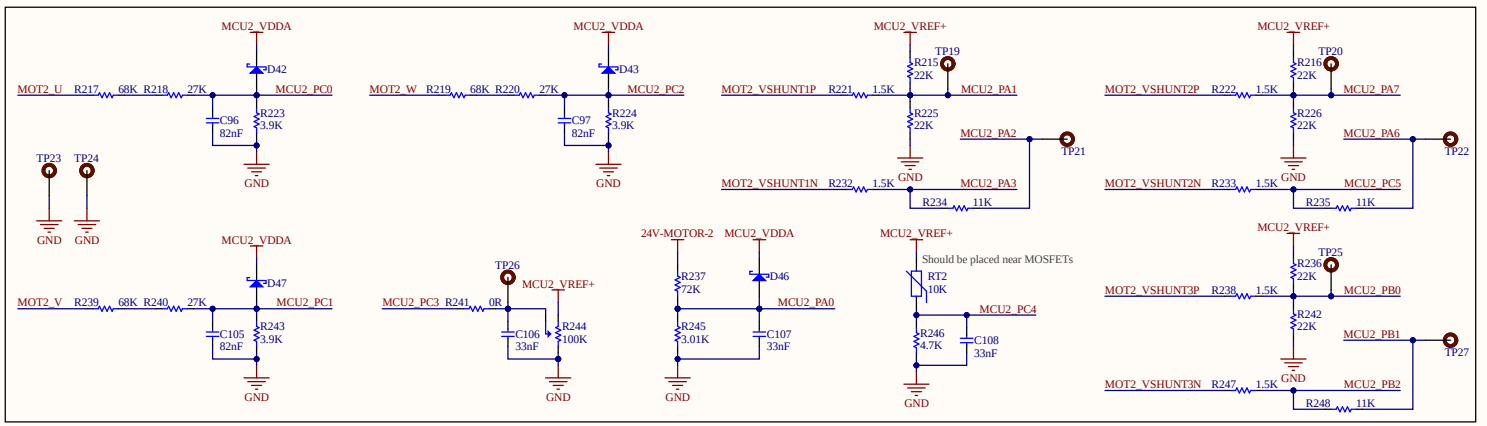
CONTROLLER WITH DRIVER

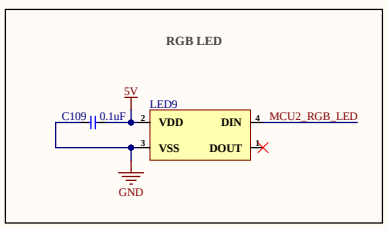












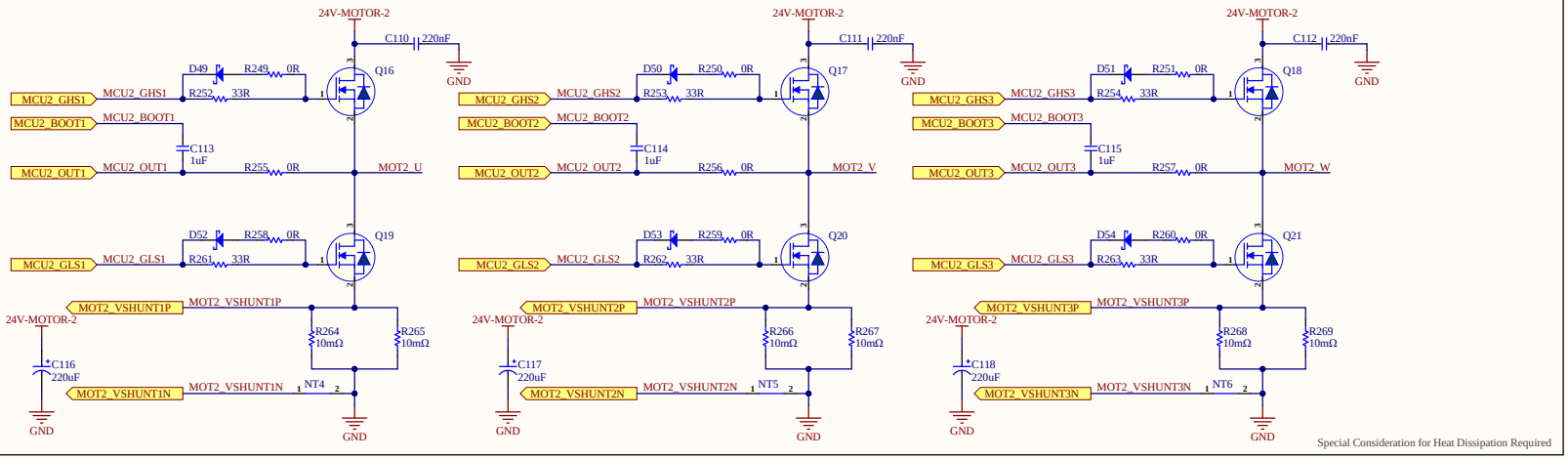
MOT2_VSHUNT1P
MOT2_VSHUNT1N
MOT2_VSHUNT1P
MOT2_VSHUNT1N
MOT2_VSHUNT2P
MOT2_VSHUNT2N
MOT2_VSHUNT2P
MOT2_VSHUNT2N
MOT2_VSHUNT3P
MOT2_VSHUNT3N
MOT2_VSHUNT3P
MOT2_VSHUNT3N

MOT2_U
MOT2_V
MOT2_W
MOT2_U
MOT2_V
MOT2_W

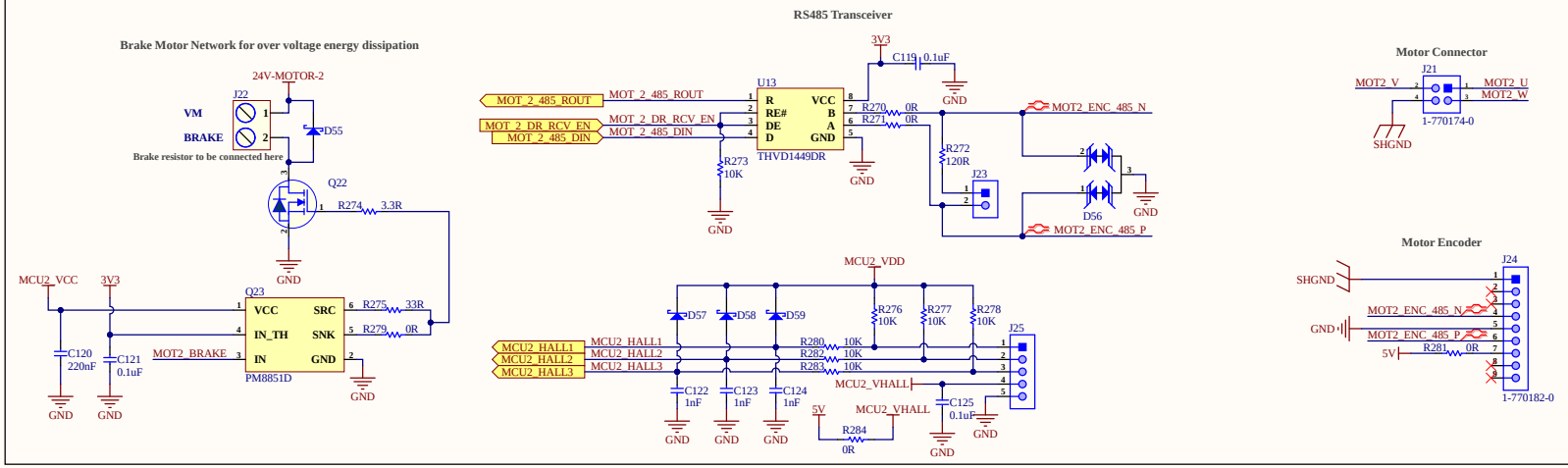
24V-MOTOR-2
5V
3V3
MCU2_VDD
MCU2_VCC
GND
24V-MOTOR-2
5V
3V3
MCU2_VDD
MCU2_VCC
GND

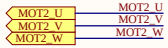
Title: Controller-Driver-Sheet-2.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS





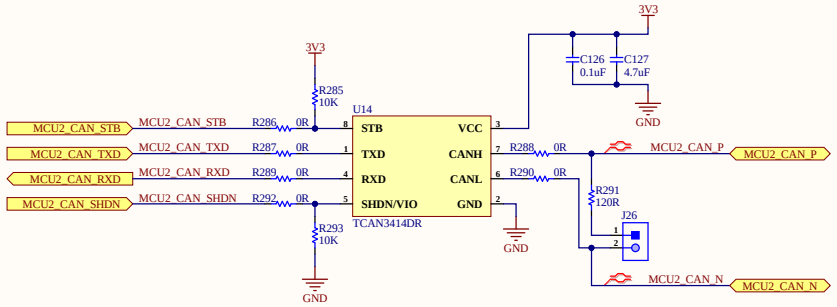




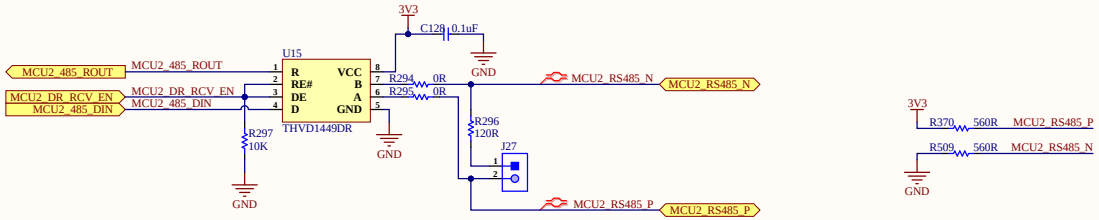
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-2.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485

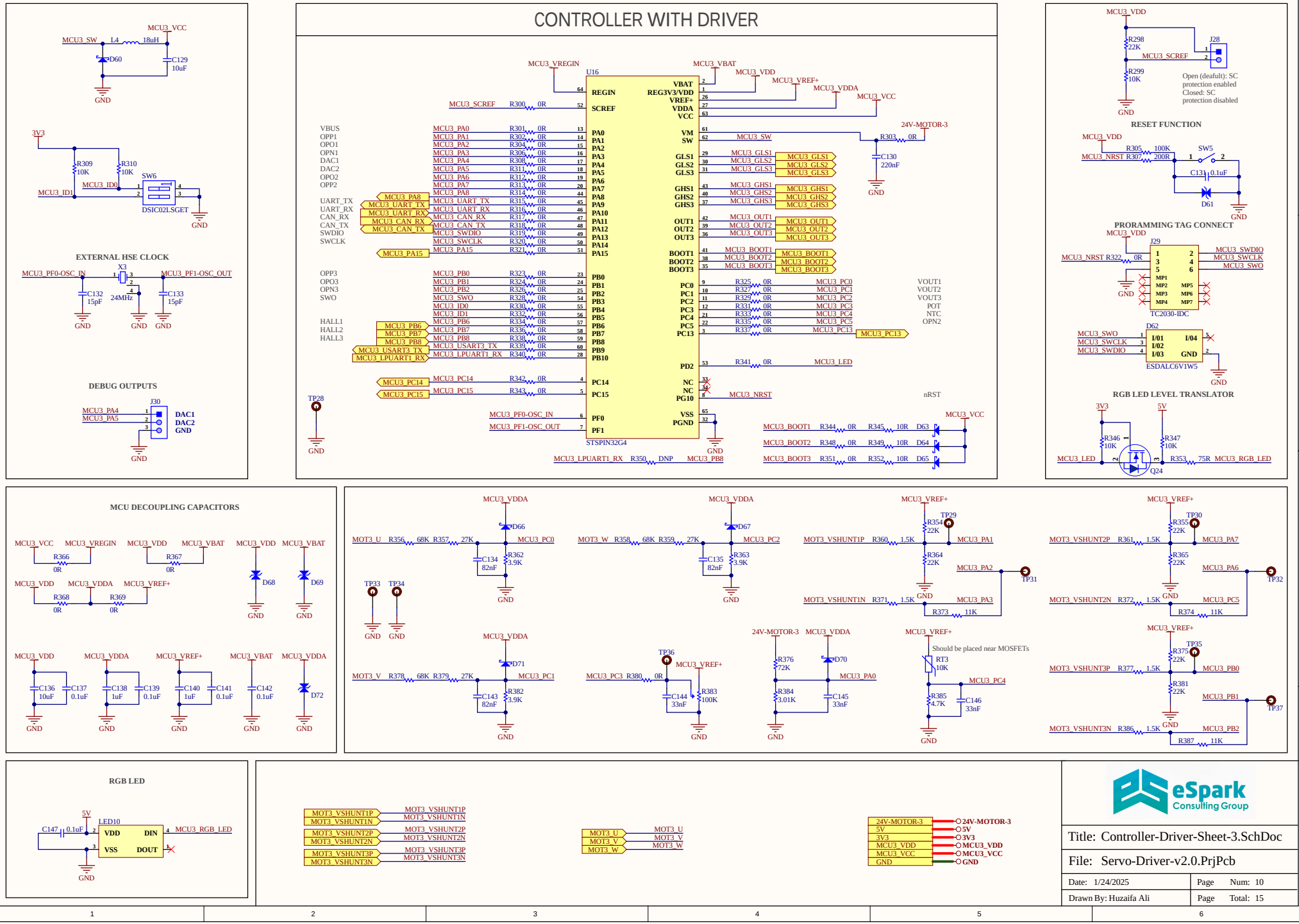


3V3 — 3V3
GND — GND

eSpark Consulting Group

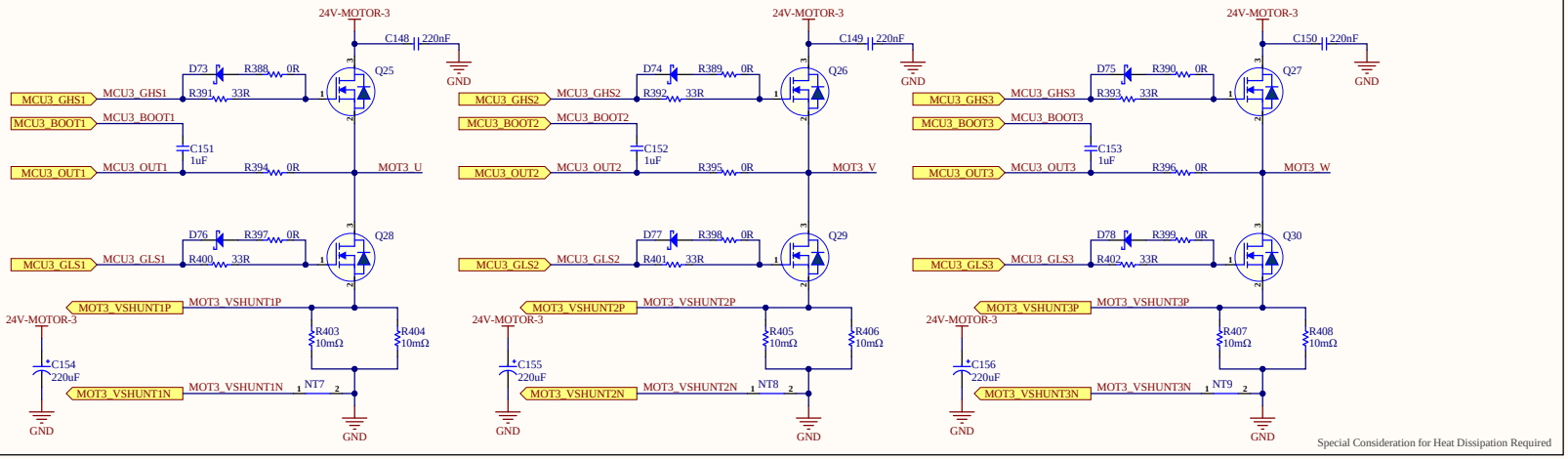
Title: Controller-Comms-Sheet-2.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/29/2025	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 15

CONTROLLER WITH DRIVER

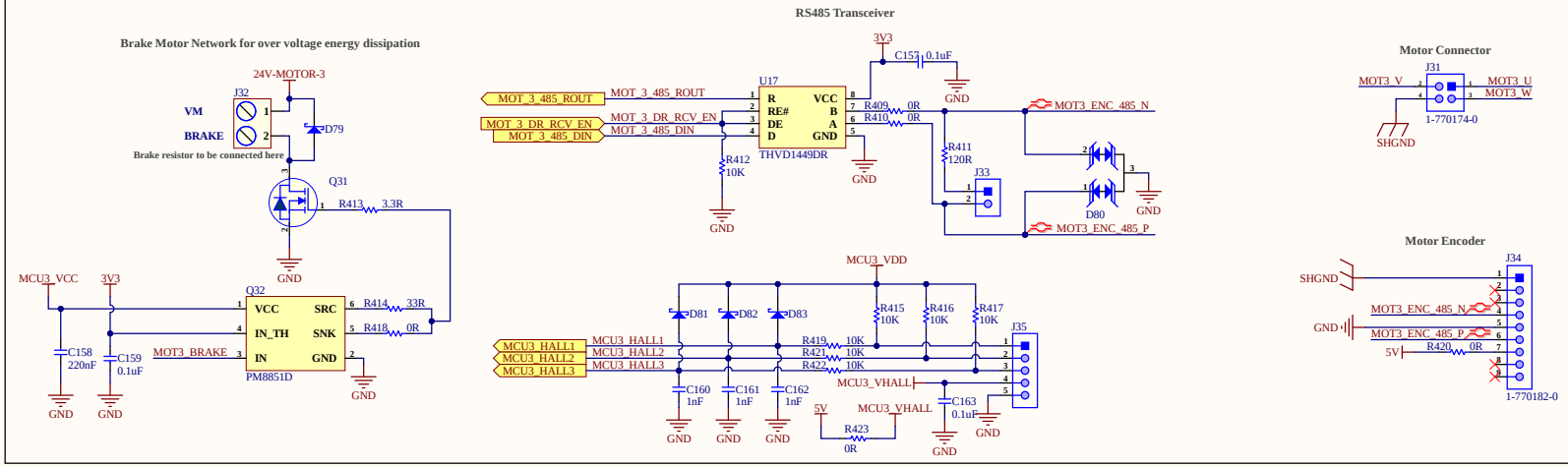


Title:	Controller-Driver-Sheet-3.SchDoc
File:	Servo-Driver-v2.0.PrjPcb
Date: 1/24/2025	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS



MOT3_U
MOT3_V
MOT3_W
MOT3_U
MOT3_V
MOT3_W

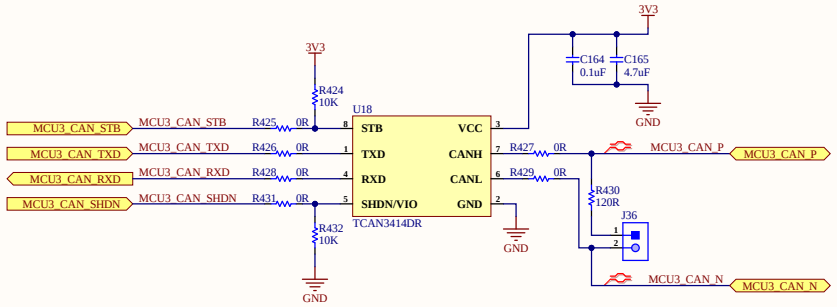
MOT3_BRAKE
MOT3_BRAKE

24V-MOTOR-3
5V
3V3
MCU3_VDD
MCU3_VCC
GND
SHGND
24V-MOTOR-3
5V
3V3
MCU3_VDD
MCU3_VCC
GND
SHGND

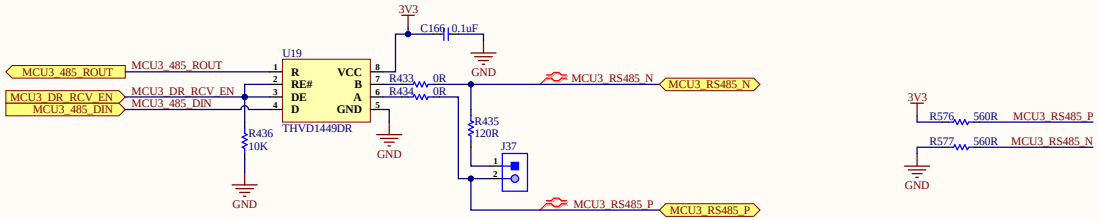
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-3.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485

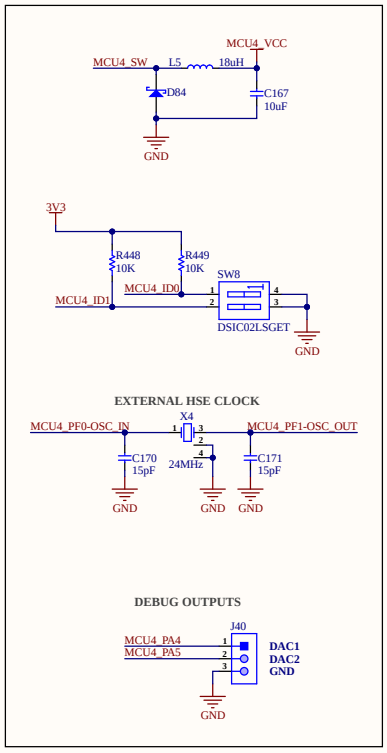


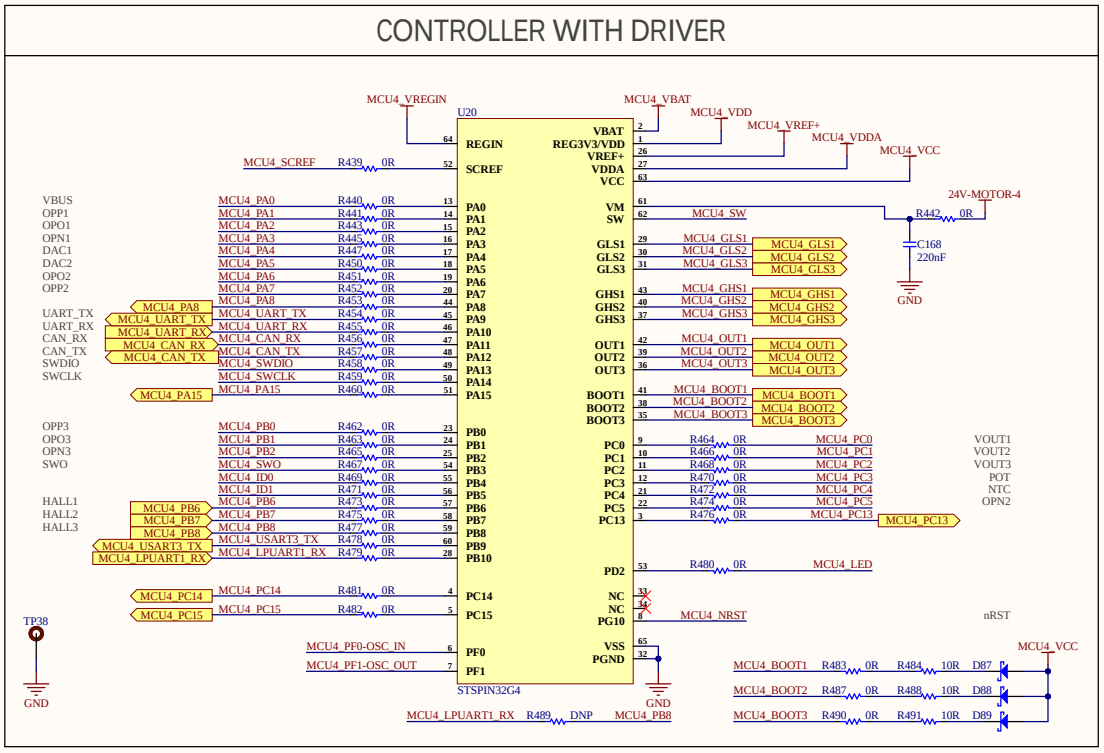
3V3 — 3V3
GND — GND

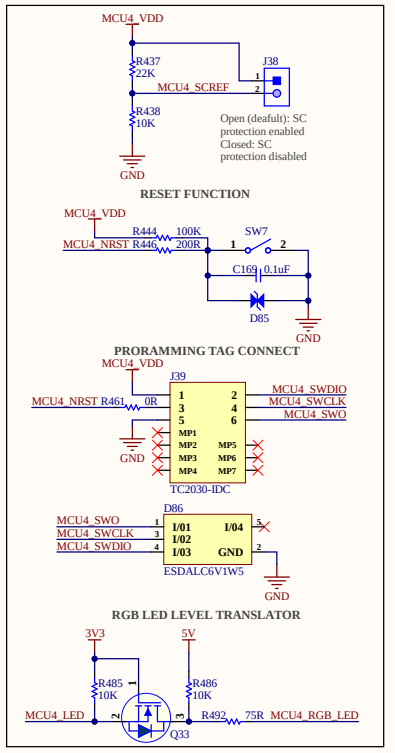
eSpark Consulting Group

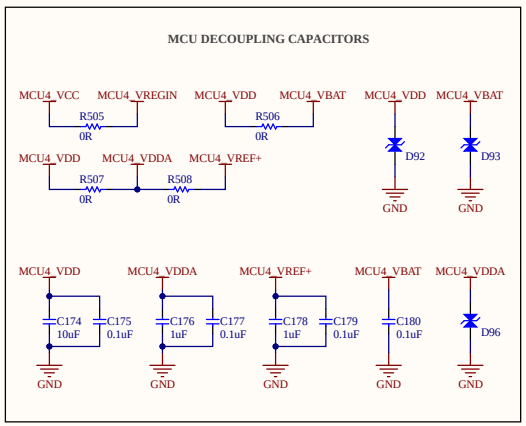
Title: Controller-Comms-Sheet-3.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/29/2025	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 15

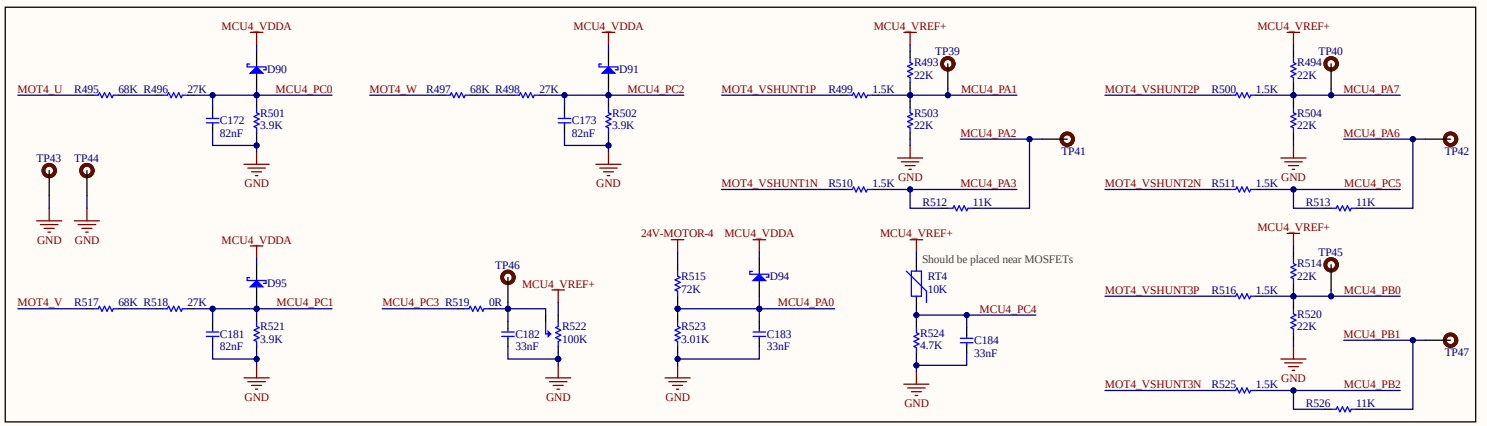
CONTROLLER WITH DRIVER

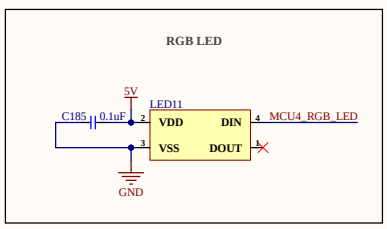












MOT4_VSHUNT1P
MOT4_VSHUNT1N
MOT4_VSHUNT2P
MOT4_VSHUNT2N
MOT4_VSHUNT3P
MOT4_VSHUNT3N

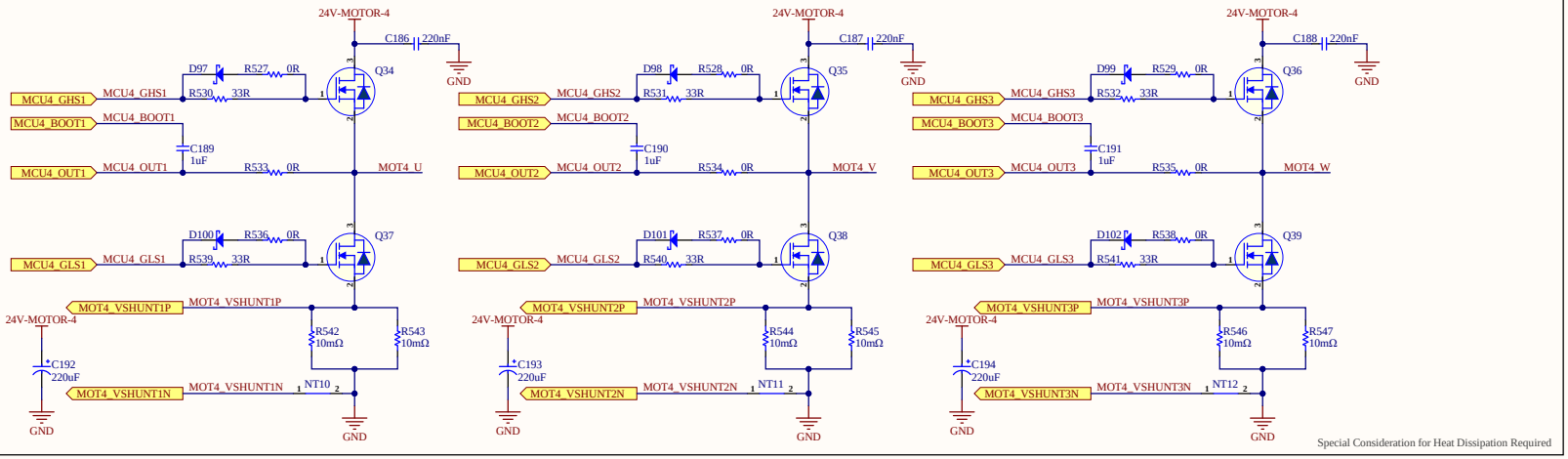
MOT4_U
MOT4_V
MOT4_W

24V-MOTOR-4
5V
3V3
MCU4_VDD
MCU4_VCC
GND

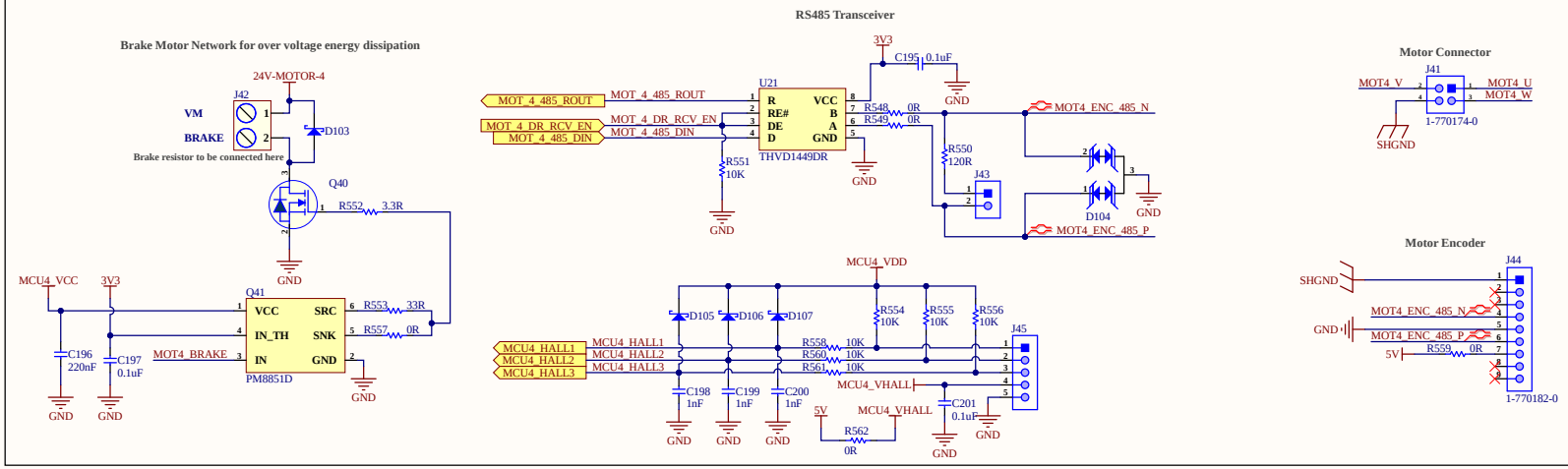
eSpark Consulting Group

Title: Controller-Driver-Sheet-4.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS



MOT4_U → MOT4_U
MOT4_V → MOT4_V
MOT4_W → MOT4_W

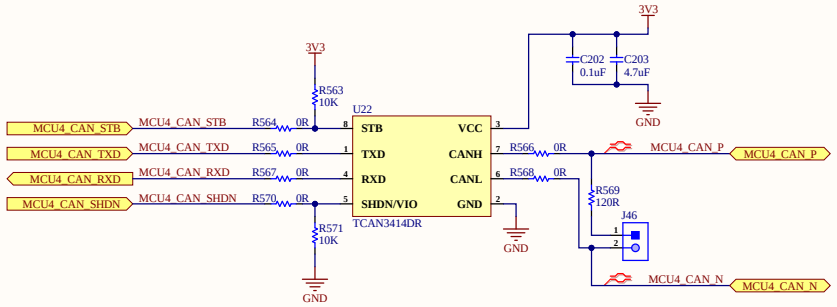
MOT4_BRAKE → MOT4_BRAKE

24V-MOTOR-4 → 24V-MOTOR-4
5V → 5V
3V3 → 3V3
MCU4_VDD → MCU4_VDD
MCU4_VCC → MCU4_VCC
GND → GND
SHGND → SHGND

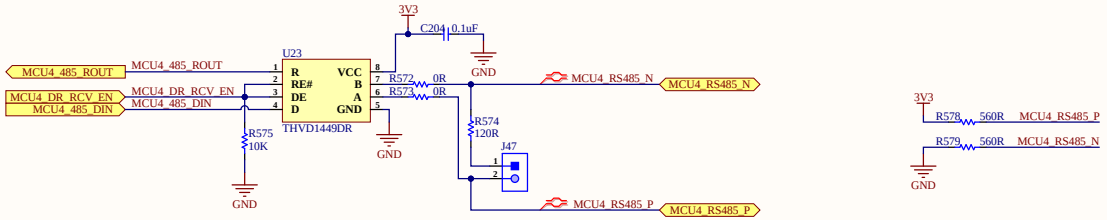
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-4.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485



3V3 — 3V3
GND — GND

eSpark Consulting Group

Title: Controller-Comms-Sheet-4.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 1/29/2025	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 15